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Public 1st place solution writeup

Node-centric LSTM with node-specific projections for rainfall-driven water level prediction

UrbanFloodBench: Flood Modelling

Solution Writeup · 1st place · Mar 16, 2026

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First of all, I would like to thank the host for organizing this competition and Kaggle for making it possible.
I initially approached this competition with a more general and complex model, but I soon realized that the water levels follow a very strong pattern driven by the cumulative sum of rainfall. The problem therefore favors a simpler solution. I simplified the approach to a straightforward **LSTM with node-specific projections**.

My key insight was that the model essentially needs to learn a few distinct patterns:

- *The temporal cumulative pattern of rainfall.*
- *How this pattern evolves based on the initial hydraulic conditions.*
- *Node-specific behavior, since while all nodes follow similar dynamics, some lack water storage systems or have different release characteristics.*

Data Processing

My modeling followed a node-centric approach. For each node, I gathered relevant edge information:

- Incoming edges: properties of edges flowing into the node.
- Outgoing edges: properties of edges flowing out from the node.

The dynamic features of edges were combined with the node's own dynamic attributes, such as water level, flow, and inlet flow, while the static features of edges were combined with the node's static attributes.

This approach was applied to both 1D and 2D nodes, allowing the model to capture the time-varying behavior and structural context across all node types.

Modeling / Architecture

The core of the solution is a node-centric LSTM model with node-specific projections for both 1D and 2D nodes. The architecture is designed to capture temporal cumulative patterns of rainfall while also accounting for node-specific hydraulic and structural characteristics.

Input Processing

- Water level is normalized relative to **minimum elevation** or **invert elevation**, ensuring consistency across nodes with different baseline heights and scales.
- For dynamic features, delta of water level is crafted, and information of delta and water level from neighboring nodes is incorporated to capture **network interactions** effectively.
- Dynamic features from the last of the first 10 timesteps are processed through **dense layers** to form the **initial hidden state** of the LSTM.
- Static features are processed through a **dense layer** to form the **initial cell state**, providing structural context for each node.
- Rainfall data is expanded and projected using **node-specific projections**, allowing the model to account for the fact that each node responds differently to the same rainfall input.

LSTM Layers

- Separate LSTM layers are used for 1D and 2D nodes, modeling their temporal dynamics independently.
- The LSTM learns the **cumulative effects of rainfall** and how they interact with the initial hydraulic conditions.

Node-specific Projection

- After the LSTM, a **node-specific projection layer** maps the LSTM outputs to the final predicted values.
- This projection accounts for **heterogeneity across nodes**, including differences in storage, release behavior, and connectivity.

Training Method

Prediction Targets & Loss Function

- For both 1D and 2D nodes, the model predicts the **delta of water volume**. Within the model, the base area of 1D nodes and the surface area of 2D nodes are used to convert these deltas into water level changes.
- Both node types are trained using **MSE loss**, but with slightly different targets:
  - **1D nodes:** MSE is computed on the **reconstructed water level**, obtained by cumulatively summing the predicted deltas.
  - **2D nodes:** MSE is computed directly on the **predicted deltas**.

Optimizer & Training Strategy

- The model is trained using **Nadam optimizer** combined with **Lookahead**.
- Nadam hyperparameters: β_1 = 0.99, β_2 = 0.999, ε = 1e-7.
- Lookahead hyperparameters: **synchronization period = 6**, **sync alpha = 0.5**.
- Scheduler: Cosine decay
- Epochs: 6000
- Batch size: 8

Post-Processing

During inference, the reconstructed water levels for each node (obtained from the predicted deltas) are **clipped between the minimum and maximum water levels** observed in the training data.
This ensures that predictions remain **physically plausible** and consistent with historical observations.

Results

For the final submission, the model was trained on the full dataset. The best performance was obtained using an ensemble of 5 models (trained with different random seeds) for Urban Model 2, while Urban Model 1 used a single model.

Number of Models	Public LB
Single model	0.0154
Ensemble of 2	0.0124
Ensemble of 5	0.0110

The results demonstrate that **ensembling significantly improves performance**, highlighting the benefit of combining models with different initializations to reduce variance and capture complementary patterns.

Conclusion

In this competition, a **node-centric LSTM with node-specific projections** proved effective at capturing both the temporal cumulative patterns of rainfall and the diverse hydraulic behavior across nodes. Feature normalization, neighbor information, and careful handling of dynamic and static features contributed to stable and accurate predictions. Finally, ensembling multiple models helped reduce variance and improved overall performance, securing a strong leaderboard result.

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